Scalar ModDiv Revisited

Exact Interval Inversion, Posterior Key Prediction, Three-Quarters Agreement, and Alignment Limits

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Abstract

We give a self-contained analysis of a scalar modulo-division map of the form

$$x \longmapsto \left\lfloor \frac{[ax]_P}{Q} \right\rfloor,$$

with exact Euclidean arithmetic throughout. Two phenomena that are often conflated are separated. First, for a secret known to lie in a public integer interval, compatible preimages can be counted and selected in polynomial time by a Euclidean floor-sum recurrence. For a uniform interval secret this yields an exact posterior sampler, and a replacement argument shows that a passive observer can predict the shared scalar key with exactly the honest agreement probability in the specified two-message deterministic model. Second, for the full-range power-of-two experiment with $P = 2^p$, $Q = 2^q$, $p > 2q$, uniform multiplier, and independent uniform secrets in $\{0, \ldots, Q-1\}$, we prove an agreement probability strictly above three quarters. The proof tracks the correlated remainders generated by the common multiplier, obtains an analytic bound for $q \geq 12$, and closes $q = 1, \ldots, 11$ by exact finite computation. The guarantee is distribution-specific: exactly seven-bit positive secrets give an explicit probability below three quarters. For aligned general moduli $P = JQ^2$, agreement is independent of J and tends to $41/54$ as $Q \to \infty$; a nonaligned family $P = 3Q^2/2$ has limiting agreement strictly below $3/4$. These are correctness and inversion results, not a lattice-hardness reduction or a deployment recommendation.

Keywords: modular reduction; integer division; quotient extraction; key agreement; exact inversion; posterior sampling; floor sums; correlated remainders; asymptotic agreement.

1 Public record and contribution boundary

The scalar ModDiv line was publicly described through the observation

$$F(x) = \left\lfloor \frac{[ax]_{2^p}}{2^q} \right\rfloor$$

and associated key-exchange constructions in the public record of Azhari and Bouftass [1]. Zhang later described a practical attack on the scalar key exchange using a two-dimensional closest-vector formulation and Gaussian lattice reduction [2]. Barcau, Paşol, Pleşca and Togan published a SecITC 2017 paper entitled *On a Key Exchange Protocol*, which we cite only as public historical context [3]. A later multidimensional ModDiv proposal explicitly identifies the one-dimensional construction as its predecessor [4].

Historical claims in this article are limited to the public references above. The mathematical results used below are stated and proved self-containedly, apart from standard elementary facts. We make no exhaustive novelty claim for Euclidean floor-sum algorithms, Bayes' rule, finite-group character arguments, or standard divisor estimates.

Three distinctions are essential:

(i) agreement of two honest computations is not confidentiality;

(ii) finding a compatible preimage is not necessarily recovering the originally sampled secret;

(iii) a theorem for one exact secret distribution does not transfer to a different support without proof.

2 The scalar quotient map and a precise exchange model

For a positive integer M, let $[z]_M \in \{0, \dots, M-1\}$ denote the canonical residue of z modulo M.

Definition 2.1 (Modulo-division)**.** *For positive integers P, Q, define*

$$\mathrm{MD}_{P,Q}(z) = \left\lfloor \frac{[z]_P}{Q} \right\rfloor.$$

All divisions in this paper are Euclidean floor divisions.

The interval inversion results below allow arbitrary positive integers P, Q, including $Q \nmid P$. For the two-message exchange model, take P, Q, H, D, J to be positive integers and a, L_X, R_X, L_Y, R_Y to be integers, with

$$P = QH, \qquad H = DJ,$$

and with nonempty public intervals $L_X < R_X$ and $L_Y < R_Y$. For every fixed choice of the public parameters, including a and the intervals, the private variables X, Y are independent and uniform on

$$I_X = \{L_X, \dots, R_X - 1\}, \qquad I_Y = \{L_Y, \dots, R_Y - 1\}.$$

If the public parameters themselves are random, this is a conditional-law assumption given those public parameters. The parties publish

$$U = \mathrm{MD}_{P,Q}(aX), \qquad V = \mathrm{MD}_{P,Q}(aY),$$

and compute

$$T_A = [XV]_H, \qquad T_B = [YU]_H,$$
$$K_A = \left\lfloor \frac{T_A}{D} \right\rfloor, \qquad K_B = \left\lfloor \frac{T_B}{D} \right\rfloor \in \{0, \dots, J-1\}. \tag{2.1}$$

There is no extra reconciliation message or secret-dependent confirmation value in this model.

If

$$r_x = [aX]_Q, \qquad r_y = [aY]_Q,$$

then Euclidean division gives integers k_x, k_y satisfying

$$aX = Pk_x + QU + r_x, \qquad aY = Pk_y + QV + r_y.$$

Multiplying by the opposite secrets and subtracting yields the exact discrepancy

$$XV - YU = H(Yk_x - Xk_y) + \delta, \qquad \delta = \frac{Yr_x - Xr_y}{Q} \in \mathbb{Z}. \tag{2.2}$$

Thus $T_A - T_B \equiv \delta \pmod{H}$. This identity is useful for correctness analysis, but small circular displacement alone does not determine whether the final division by D crosses a cell boundary.

Example 2.2 (Small displacement can still disagree)**.** *Take $P = 64$, $Q = 4$, $H = 16$, $D = 4$, $a = 35$, $X = 2$, $Y = 3$. Then*

$$U = 1, \quad V = 10, \quad T_A = 4, \quad T_B = 3,$$

so $K_A = 1$ and $K_B = 0$, although the intermediate discrepancy is only one. A correctness probability therefore requires the joint law of displacement and cell position.

3 Exact interval inversion

Fix public integers a, u, L, R and positive integers P, Q, with $L \le R$, and put

$$I = \{L, \dots, R-1\}, \qquad n = R - L.$$

The case $L = R$ is permitted for counting, but a uniform interval prior below requires $L < R$. The inverse problem asks for $x \in I$ such that

$$\mathrm{MD}_{P,Q}(ax) = u. \tag{3.1}$$

For arbitrary positive P, Q, define

$$\ell_u = Qu, \qquad h_u = \min\{Q(u+1), P\}.$$

If $u < 0$ or $Qu \ge P$, the inverse set is empty.

For $n \ge 0$, $m > 0$, and signed integers a, b, define the floor sum

$$S(n, m, a, b) = \sum_{i=0}^{n-1} \left\lfloor \frac{ai + b}{m} \right\rfloor. \tag{3.2}$$

Lemma 3.1 (Euclidean floor-sum recurrence)**.** *Write*

$$a = q_a m + a_0, \qquad b = q_b m + b_0, \quad 0 \le a_0, b_0 < m.$$

Then

$$S(n, m, a, b) = q_a \frac{n(n-1)}{2} + q_b n + S(n, m, a_0, b_0). \tag{3.3}$$

If $0 < a < m$, $0 \le b < m$, and $an + b = mT + c$ with $0 \le c < m$, then

$$S(n, m, a, b) = S(T, a, m, c). \tag{3.4}$$

The recurrence terminates after $O(1 + \log m)$ nontrivial Euclidean reductions and has polynomial bit complexity in the binary input length.

Proof. Equation (3.3) follows by separating the integral multiples of m inside each floor. For (3.4), count lattice points by horizontal level. The number of indices $0 \le i < n$ satisfying $ai + b \ge mj$, for $1 \le j \le T$, equals

$$n - \left\lceil \frac{mj - b}{a} \right\rceil = \left\lfloor \frac{an + b - mj}{a} \right\rfloor.$$

Summing over j and substituting $an + b = mT + c$ gives (3.4). After normalization, the new modulus is $a < m$; subsequent moduli follow Euclidean remainders. Integer sizes remain polynomial in the input bit length. □

Theorem 3.2 (Exact interval counting and ranked selection)**.** *The number $N_I(a, u)$ of $x \in I$ satisfying (3.1) is*

$$N_I(a, u) = \begin{cases} 0, & u < 0 \text{ or } Qu \ge P, \\ S(n, P, a, aL - \ell_u) - S(n, P, a, aL - h_u), & 0 \le Qu < P. \end{cases} \tag{3.5}$$

It can be computed in polynomial time in the binary input length. If $N_I(a, u) > 0$, the k-th compatible secret in increasing order can be selected in polynomial time for every $1 \le k \le N_I(a, u)$.

Proof. For every integer w and $0 \le \ell \le h \le P$,

$$\mathbf{1}\{\ell \le [w]_P < h\} = \left\lfloor \frac{w - \ell}{P} \right\rfloor - \left\lfloor \frac{w - h}{P} \right\rfloor. \tag{3.6}$$

Indeed, writing $w = Pj + r$ with $0 \le r < P$, the two floors differ by one exactly when $r \in [\ell, h)$. Substitute $w = a(L + i)$ and sum over $0 \le i < n$ to obtain (3.5).

For ranked selection, bisect $[L, R)$, count compatible points in the left half, and retain the half containing the requested rank. At most $\lceil \log_2 n \rceil$ bisections are needed; each invokes the polynomial-time counter. □

The theorem concerns the scalar interval-constrained inverse relation. It does not imply that a multidimensional rounded system can be inverted by the same method.

Corollary 3.3 (Exact posterior sampling)**.** *Assume $L < R$. If X is uniform on I, then conditional on $U = \mathrm{MD}_{P,Q}(aX) = u$, the posterior distribution of X is uniform on the $N_I(a, u)$ compatible integers. It therefore has an expected-polynomial-time exact sampler whenever the event has positive probability.*

Proof. The prior mass of every point of I is equal and publication is deterministic, so Bayes' rule gives equal posterior mass to every compatible point. Sample a uniform rank in $\{1, \ldots, N_I(a, u)\}$ and apply ranked selection. Rejection sampling a uniform rank from fair bits succeeds with probability greater than $1/2$ per trial unless $N_I = 1$. □

A cap of t rank-generation trials gives strict polynomial running time with failure probability at most 2^{-t}; any abort or fallback must be included in the security accounting.

4 Posterior replacement predicts the shared key

Let the complete public transcript be

$$\mathcal{T} = (\text{public parameters}, a, U, V).$$

Allow arbitrary efficiently computable deterministic local extraction maps

$$K_A = \mathcal{A}(\mathcal{T}, X), \qquad K_B = \mathcal{B}(\mathcal{T}, Y),$$

where the interval secrets are independent and uniform conditionally on the public parameters, as specified in Section 2, and there is no additional private-state-dependent message. Put

$$\varepsilon_{\mathrm{agr}} = \Pr[K_A \neq K_B].$$

Theorem 4.1 (Posterior replacement)**.** *Given only $\mathcal{T}$, a passive observer can use fresh random coins to sample X^* from the exact posterior of X given the public parameters, a, and U, and output*

$$K^* = \mathcal{A}(\mathcal{T}, X^*).$$

Then

$$\Pr[K^* = K_B] = 1 - \varepsilon_{\mathrm{agr}}. \tag{4.1}$$

The expected running time is polynomial in the binary input length.

Proof. Condition on the public parameters and a. The variables X, Y are independent, U is a deterministic function of X, and V a deterministic function of Y. After conditioning additionally on U, the posterior of X is independent of (Y, V). The previous corollary samples exactly that posterior, hence

$$(\mathcal{T}, Y, X^*) \overset{d}{=} (\mathcal{T}, Y, X).$$

Applying the same event $\mathcal{A}(\mathcal{T}, \cdot) = \mathcal{B}(\mathcal{T}, Y)$ to both joint laws proves (4.1). □

This does not claim that every compatible preimage yields the same key, nor that $X^* = X$. It is a distributional replacement statement.

Corollary 4.2 (Real-key versus uniform-key test)**.** *If the key alphabet has fixed public size J, equality with K^* has real-minus-independent-uniform acceptance gap*

$$(1 - \varepsilon_{\mathrm{agr}}) - \frac{1}{J}. \tag{4.2}$$

Under a challenge-bit convention where advantage is measured as success bias above one half, the corresponding bias is half of the acceptance-probability gap.

Additional reconciliation or confirmation messages depending on the original private state can change the posterior and are outside this theorem.

5 The full-range power-of-two agreement experiment

Let $p, q \in \mathbb{Z}$ with $q \geq 1$ and $p > 2q$, and put

$$Q = 2^q, \qquad P = 2^p, \qquad J = P/Q^2 \geq 2, \qquad H = P/Q = QJ. \tag{5.1}$$

Sample independently

$$A \leftarrow \mathrm{Unif}\{0, \ldots, P-1\}, \qquad X, Y \leftarrow \mathrm{Unif}\{0, \ldots, Q-1\}. \tag{5.2}$$

Publish

$$U = \left\lfloor \frac{[AX]_P}{Q} \right\rfloor, \qquad V = \left\lfloor \frac{[AY]_P}{Q} \right\rfloor, \tag{5.3}$$

and compute

$$W_A = \left\lfloor \frac{[QXV]_P}{Q^2} \right\rfloor, \qquad W_B = \left\lfloor \frac{[QYU]_P}{Q^2} \right\rfloor. \tag{5.4}$$

Equivalently, because $[Qz]_P = Q[z]_H$,

$$W_A = \left\lfloor \frac{[XV]_H}{Q} \right\rfloor, \qquad W_B = \left\lfloor \frac{[YU]_H}{Q} \right\rfloor. \tag{5.5}$$

Write $\alpha_q = \Pr[W_A = W_B]$.

The support in (5.2) includes zero. It is not the set of positive integers having exactly q bits.

Decompose

$$A = a + Qh + Q^2 z, \qquad 0 \leq a, h < Q, \quad 0 \leq z < J. \tag{5.6}$$

The three digits are independent uniforms. For fixed a, x, y, define

$$f_a(x) = \left\lfloor \frac{ax}{Q} \right\rfloor, \qquad \rho_a(x) = [ax]_Q,$$

$$u = x f_a(y), \qquad v = y f_a(x), \qquad \Delta = v - u. \tag{5.7}$$

Theorem 5.1 (Exact two-digit agreement event)**.** *One has*

$$Q\Delta = x\rho_a(y) - y\rho_a(x), \qquad |\Delta| < Q, \tag{5.8}$$

and, with

$$T = [xyh + u]_Q,$$

$$W_A = W_B \quad \Longleftrightarrow \quad 0 \leq T + \Delta < Q. \tag{5.9}$$

Consequently α_q is independent of the higher public precision p as long as $p > 2q$.

Proof. Expanding $ax = Qf_a(x) + \rho_a(x)$ and $ay = Qf_a(y) + \rho_a(y)$ gives (5.8), and the numerator has absolute value at most $(Q-1)^2 < Q^2$. Reducing the publications modulo H yields

$$U \equiv f_a(x) + hx + Qxz \pmod{H}, \qquad V \equiv f_a(y) + hy + Qyz \pmod{H}.$$

Thus $YU - XV \equiv \Delta \pmod{H}$ and $[XV]_Q = [xyh + u]_Q$. Writing $[XV]_H = QW_A + T$, the quotient changes by zero exactly when $0 \leq T + \Delta < Q$. The event depends on a, h, x, y but not on z or J. □

The theorem tracks the actual common-multiplier dependence. It does not replace $\rho_a(X)$ and $\rho_a(Y)$ by independent errors.

6 Exact phase counting and the analytic three-quarters bound

Fix a, x, y and put

$$g = \gcd(xy, Q), \qquad m = Q/g,$$

with $\gcd(0, Q) = Q$. As h varies uniformly, $T = [xyh + u]_Q$ is uniform on one residue class modulo g.

Proposition 6.1 (Conditional disagreement count)**.** *Under the full-range experiment,*

$$\Pr[W_A \neq W_B \mid a, x, y] = \frac{g}{Q}\left|\left\lfloor \frac{v}{g} \right\rfloor - \left\lfloor \frac{u}{g} \right\rfloor\right|. \tag{6.1}$$

Moreover,

$$|g\,|\lfloor v/g \rfloor - \lfloor u/g \rfloor| - |v - u|| \leq g - 1. \tag{6.2}$$

Proof. Write $u = gU_0 + s$ and $v = gV_0 + r$, $0 \leq s, r < g$. The possible phases are $s + gj$, $0 \leq j < m$. Since

$$T + \Delta = g(j + V_0 - U_0) + r,$$

agreement is equivalent to $0 \leq j + c < m$ with $c = V_0 - U_0$. Exactly $|c|$ of the m indices fail. Equation (6.2) follows from $v - u = gc + r - s$ and the reverse triangle inequality. □

Define

$$\mathcal{D}_Q = \frac{1}{Q^2}\mathbb{E}_{a,X,Y}\left|X\rho_a(Y) - Y\rho_a(X)\right|. \tag{6.3}$$

Averaging (6.2) gives

$$1 - \alpha_q \leq \mathcal{D}_Q + \frac{\mathbb{E}g - 1}{Q}. \tag{6.4}$$

Lemma 6.2 (Two-adic gcd averages)**.** *For independent uniform $X, Y \in \{0, \ldots, 2^q - 1\}$,*

$$\mathbb{E}\gcd(XY, 2^q) = 1 + \frac{q(q+5)}{8}, \tag{6.5}$$

and

$$\mathbb{E}[\gcd(X, 2^q)\mathbf{1}_{\{X>0\}}] = \frac{q}{2}. \tag{6.6}$$

Proof. For $1 \leq j \leq q$, splitting according to the two-adic valuation of X gives

$$\Pr[2^j \mid XY] = \frac{j+2}{2^{j+1}}.$$

Since a power-of-two gcd satisfies

$$\gcd(XY, 2^q) = 1 + \sum_{j=1}^{q} 2^{j-1}\mathbf{1}_{\{2^j \mid XY\}},$$

expectation yields (6.5). For (6.6), exactly $Q/2^{i+1}$ nonzero residues below Q have valuation i, for $0 \leq i < q$. □

Hence

$$1 - \alpha_q \leq \mathcal{D}_Q + \frac{q(q+5)}{8Q}. \tag{6.7}$$

To bound $\mathcal{D}_Q$, introduce an auxiliary independent multiplier b uniform modulo Q and set

$$\mathcal{D}_Q^{\mathrm{ind}} = \frac{1}{Q^2}\mathbb{E}_{a,b,X,Y}\left|X\rho_b(Y) - Y\rho_a(X)\right|.$$

Lemma 6.3 (Variance decoupling)**.**

$$\mathcal{D}_Q \leq \mathcal{D}_Q^{\mathrm{ind}}. \tag{6.8}$$

Proof. For

$$G_a(t) = \frac{1}{Q}\sum_{x=1}^{Q-1} x\mathbf{1}_{\{\rho_a(x)>tx\}},$$

use $|r-s| = r+s-2\min(r,s)$ and

$$\min(x\rho_a(y), y\rho_a(x)) = xy\int_0^\infty \mathbf{1}_{\{\rho_a(x)>tx\}}\mathbf{1}_{\{\rho_a(y)>ty\}}\,dt.$$

Averaging gives

$$\mathcal{D}_Q^{\mathrm{ind}} - \mathcal{D}_Q = \frac{2}{Q^2}\int_0^\infty \mathrm{Var}_a(G_a(t))\,dt \geq 0.$$

All integrals are finite sums of step-function integrals. □

For $x > 0$, put $d_x = \gcd(x, Q)$. The residue $\rho_a(x)$ is uniform on $\{0, d_x, \ldots, Q-d_x\}$. If R is continuous uniform on $[0, Q)$, then $d_x\lfloor R/d_x\rfloor$ has exactly this distribution and mean rounding loss $d_x/2$. With independent continuous uniforms ξ, η on $[0,1]$, define

$$\mathcal{J}_Q = \mathbb{E}\left|\frac{X}{Q}\eta - \frac{Y}{Q}\xi\right|.$$

The rounding comparison and (6.6) give

$$\mathcal{D}_Q^{\mathrm{ind}} \leq \mathcal{J}_Q + \frac{q(Q-1)}{4Q^2}. \tag{6.9}$$

Lemma 6.4 (Exact continuous-discrete average)**.** *For every integer $Q \geq 2$,*

$$\mathcal{J}_Q = \frac{13}{54} - \frac{1}{9Q} - \frac{1}{54Q^2} - \frac{1}{9Q^3} < \frac{13}{54}. \tag{6.10}$$

Proof. For $0 \leq x \leq y$, $y > 0$,

$$\int_0^1\int_0^1 \left|\frac{x}{Q}t - \frac{y}{Q}s\right| ds\,dt = \frac{y-x}{2Q} + \frac{x^2}{3Qy}.$$

Sum the diagonal and both off-diagonal halves over $0 \leq x, y < Q$. The result simplifies to

$$\frac{1}{Q^3}\sum_{y=1}^{Q-1}\left(\frac{13y^2}{18} + \frac{y}{2} + \frac{1}{9}\right),$$

and the elementary formulas for $\sum y$ and $\sum y^2$ give (6.10). □

Combining (6.8)–(6.10) yields the convenient bound

$$\mathcal{D}_Q \leq \frac{13}{54} + \frac{q}{4Q}. \tag{6.11}$$

Theorem 6.5 (Analytic agreement bound)**.** *For every $q \geq 1$,*

$$\alpha_q \geq \frac{41}{54} - \frac{q(q+7)}{8\,2^q}. \tag{6.12}$$

For $q \geq 12$,

$$\alpha_q \geq c_* := \frac{166397}{221184} = \frac{3}{4} + \frac{509}{221184} > \frac{3}{4}. \tag{6.13}$$

Proof. Add (6.7) and (6.11). For $q \geq 2$, $q(q+7)/2^q$ decreases because

$$2q(q+7) - (q+1)(q+8) = q^2 + 5q - 8 > 0.$$

At $q = 12$ the correction is $57/8192$, and $41/54 - 57/8192 = 166397/221184$. □

7 Finite closure and a support-sensitive counterexample

For the remaining $q = 1, \ldots, 11$, Proposition 6.1 gives the exact finite formula

$$N_{\mathrm{err}}(q) = \sum_{a,x,y=0}^{Q-1} g_{x,y} \left| \left\lfloor \frac{y\lfloor ax/Q \rfloor}{g_{x,y}} \right\rfloor - \left\lfloor \frac{x\lfloor ay/Q \rfloor}{g_{x,y}} \right\rfloor \right|, \quad g_{x,y} = \gcd(xy, Q), \tag{7.1}$$

with

$$\alpha_q = 1 - \frac{N_{\mathrm{err}}(q)}{Q^4}. \tag{7.2}$$

The supplied exact counter evaluates this formula with integer arithmetic.

q	exact α_q	$4N_{\mathrm{agree}} - 3Q^4$
1	1	16
2	123/128	216
3	459/512	2,400
4	6901/8192	24,224
5	52809/65536	234,048
6	3289931/4194304	2,307,248
7	51841515/67108864	24,157,872
8	822654211/1073741824	277,565,488
9	3276319387/4294967296	3,526,010,560
10	209209404217/274877906944	48,815,584,144
11	3343422968633/4398046511104	718,209,364,880

Theorem 7.1 (Uniform three-quarters bound)**.** *For every integer $q \geq 1$ and $p > 2q$, under exactly (5.1)–(5.4),*

$$\Pr[W_A = W_B] = \alpha_q \geq c_* > \frac{3}{4}. \tag{7.3}$$

Proof. The analytic theorem covers $q \geq 12$. Exact evaluation of (7.1) gives the eleven rows above, and integer cross-multiplication verifies every row is at least c_*. The proof is therefore computer-assisted only in this finite closure; the infinite range is analytic. □

The theorem is not a negligible-failure statement and does not assert monotonicity of α_q.

Proposition 7.2 (Exactly seven-bit positive secrets can fall below three quarters)**.** *Set*

$$q = 7, \quad p = 15, \quad Q = 128, \quad P = 32768,$$

keep A uniform on $\{0, \ldots, P-1\}$, but draw X, Y independently and uniformly from $\{64, \ldots, 127\}$. Then

$$\Pr[W_A = W_B] = \frac{12540509}{16777216} = \frac{3}{4} - \frac{42403}{16777216} < \frac{3}{4}. \tag{7.4}$$

Proof. Restricting the exact count (7.1) to the upper-half secret support gives the displayed rational value. An independent direct enumeration of the original publication and extraction formulas over all

$$32768 \cdot 64^2 = 134217728$$

triples gives exactly 100324072 agreements, whose reduced fraction is (7.4). □

Thus the three-quarters theorem is a theorem about the full-range law $\{0, \ldots, Q-1\}$, not a statement about every interval of the same bit width.

8 Agreement and passive prediction coexist

The full-range experiment is an instance of the interval model with $I_X = I_Y = \{0, \dots, Q-1\}$ and final divisor $D = Q$. Therefore the posterior-replacement theorem applies to the same experiment.

Corollary 8.1 (The agreement theorem is not secrecy)**.** *A passive expected-polynomial-time predictor W^* exists such that*

$$\Pr[W^* = W_B] = \alpha_q \geq c_*. \tag{8.1}$$

Against an independent uniform challenge key in the J-element output alphabet, the equality test has acceptance gap

$$\alpha_q - \frac{1}{J} \geq c_* - \frac{1}{2} > \frac{1}{4}. \tag{8.2}$$

This conclusion concerns exactly the deterministic two-message scalar model above. It does not automatically apply to polynomial, noisy, or reconciliation-augmented constructions.

9 Aligned general moduli and the $41/54$ limit

Now let $Q \geq 2$ and $J \geq 2$ be arbitrary integers and set

$$P = JQ^2. \tag{9.1}$$

Sample A uniformly modulo P and X, Y independently uniformly modulo Q, and keep the public and final formulas (5.3)–(5.4), with P, Q no longer required to be powers of two. Write the agreement probability as $\alpha(P, Q)$.

Decompose

$$A = a + Qh + Q^2 z, \qquad 0 \leq a, h < Q, \quad 0 \leq z < J.$$

Define $f_a, \rho_a, u, v, \Delta$ as in (5.7).

Theorem 9.1 (Alignment removes the upper multiplier)**.** *Under (9.1),*

$$Q\Delta = x\rho_a(y) - y\rho_a(x), \qquad |\Delta| < Q,$$

and

$$W_A = W_B \iff 0 \leq [xyh + u]_Q + \Delta < Q. \tag{9.2}$$

Hence $\alpha(JQ^2, Q)$ is independent of $J \geq 2$; denote the common value by α_Q.

Proof. The calculation is identical to the two-digit proof above and uses only the exact alignment $P = JQ^2$, not primality or a power-of-two modulus. The event depends on (a, h, x, y) but not on z or J. □

The conditional count (6.1)–(6.2) remains valid with general Q. Let

$$G_2(Q) = \mathbb{E}_{X,Y \bmod Q} \gcd(XY, Q).$$

If $Q = \prod_{\ell^e \| Q} \ell^e$, elementary valuation counting gives

$$G_2(Q) = \prod_{\ell^e \| Q} \left(1 + e\left(1 - \frac{1}{\ell}\right) + \frac{e(e+1)}{2}\left(1 - \frac{1}{\ell}\right)^2\right). \tag{9.3}$$

In particular $G_2(Q) = Q^{o(1)}$, so $(G_2(Q) - 1)/Q \to 0$.

Lemma 9.2 (Weak equidistribution)**.** *With independent a, X, Y uniform modulo Q,*

$$Z_Q = \left(\frac{X}{Q}, \frac{Y}{Q}, \frac{[aX]_Q}{Q}, \frac{[aY]_Q}{Q}\right)$$

converges weakly, as integer $Q \to \infty$, to four independent uniforms on $[0, 1]$.

Proof. View Z_Q on the four-torus. For fixed integers (i, j, k, l), finite character orthogonality gives

$$\mathbb{E}e^{2\pi i(iZ_1+jZ_2+kZ_3+lZ_4)} = \frac{1}{Q}\#\{a \bmod Q : \ i+ka \equiv 0, \ j+la \equiv 0 \pmod{Q}\}.$$

For every fixed nonzero character this tends to zero: if $k \neq 0$ there are at most $|k|$ solutions, if $k = 0, l \neq 0$ there are at most $|l|$, and if $k = l = 0$ a fixed nonzero i or j makes the coefficient eventually zero. The character criterion therefore gives convergence to Haar measure on the torus; passing to canonical representatives changes only boundary sets of limiting measure zero. This is weak convergence, not finite-parameter independence or total-variation convergence. A standard character-density argument is reviewed, for example, in [5]. □

Theorem 9.3 (Exact aligned asymptotic agreement)**.** *Along every sequence $Q \to \infty$ and every integer-valued $J(Q) \geq 2$,*

$$\lim_{Q\to\infty} \alpha(J(Q)Q^2, Q) = \frac{41}{54}. \tag{9.4}$$

Proof. The grid-count estimate gives

$$|1 - \alpha_Q - \mathcal{D}_Q| \leq \frac{G_2(Q) - 1}{Q} \longrightarrow 0,$$

where now

$$\mathcal{D}_Q = \frac{1}{Q^2}\mathbb{E}|X\rho_a(Y) - Y\rho_a(X)|.$$

By weak equidistribution, $\mathcal{D}_Q$ converges to $\mathbb{E}|Z_1 - Z_2|$, where Z_1, Z_2 are independent products of two independent uniforms on $[0, 1]$. For $0 < t < 1$,

$$\Pr[Z_1 > t] = 1 - t + t\log t, \qquad \mathbb{E}Z_1 = \frac{1}{4}.$$

Hence

$$\mathbb{E}\min(Z_1, Z_2) = \int_0^1 (1 - t + t\log t)^2 dt = \frac{7}{54},$$

and therefore

$$\mathbb{E}|Z_1 - Z_2| = \frac{1}{2} - 2 \cdot \frac{7}{54} = \frac{13}{54}.$$

Thus $1 - \alpha_Q \to 13/54$ and $\alpha_Q \to 41/54$. Alignment makes the result independent of $J(Q)$. □

Since $41/54 - 3/4 = 1/108 > 0$, every sufficiently large aligned integer divisor has agreement above three quarters. This statement is asymptotic; it gives neither monotonicity nor an explicit cutoff for all composite Q.

10 Nonalignment can destroy the three-quarters law

Alignment by Q^2 is not cosmetic. Let $s \geq 1$ be an integer and set

$$Q = 2^s, \qquad P = \frac{3}{2}Q^2, \qquad H = P/Q = \frac{3}{2}Q. \tag{10.1}$$

Sample A uniformly modulo P and X, Y independently uniformly modulo Q, and define the publications and outputs by the same formulas (5.3)–(5.4); write this agreement probability as $\alpha(P, Q)$. Here $Q \mid P$ but $Q^2 \nmid P$. The final quotient has the two-symbol alphabet $\{0, 1\}$, with cells of lengths Q and $Q/2$ on the circle of length H. In particular, $H/Q = 3/2$ is not the integer J of the aligned model.

Theorem 10.1 (A nonaligned family below three quarters)**.** *For the family (10.1), the limit*

$$\beta = \lim_{s\to\infty} \alpha(3Q^2/2, Q)$$

exists and satisfies

$$\beta = 1 - \frac{4}{3}\mathbb{E}\min(|Z_1 - Z_2|, 1/2) \le \frac{83}{162} + \frac{\log 2}{3} < \frac{241}{324} < \frac{3}{4}. \tag{10.2}$$

Consequently all sufficiently large members of this family violate the three-quarters bound.

Proof. For a fixed displacement $|\delta| < Q$, the disagreement of a continuous uniform phase on the circle of length $3Q/2$ is the normalized symmetric difference of the short $Q/2$ cell and its translate, namely

$$\frac{4}{3}\min\left(\frac{|\delta|}{Q}, \frac{1}{2}\right).$$

Write $A = a + Qh$, where a is uniform modulo Q and h is uniform modulo $H = 3Q/2$; these digits are independent because $P = QH$. For fixed a, x, y, the reachable phase $[xyh + u]_H$ is uniform on a residue class of step

$$g_H = \gcd(xy, H).$$

Its disagreement probability differs from the continuous phase probability by at most $4g_H/H$. Since X, Y are uniform on $[0, Q)$ and $[0, Q)^2 \subset [0, H)^2$, positivity gives

$$\mathbb{E}[g_H] \le \frac{H^2}{Q^2} G_2(H) = \frac{9}{4} G_2(H).$$

Consequently

$$\frac{4\mathbb{E}[g_H]}{H} \le \frac{9G_2(H)}{H} \longrightarrow 0,$$

by the same divisor estimate used above. Weak equidistribution then gives the displayed limit formula.

To prove the strict upper bound, use

$$\mathbb{E}(Z_1 - 1/2)_+ = -\frac{1}{16} + \frac{\log 2}{8}$$

and

$$(|z_1 - z_2| - 1/2)_+ \le (z_1 - 1/2)_+ + (z_2 - 1/2)_+.$$

Together with $\mathbb{E}|Z_1 - Z_2| = 13/54$, this yields

$$\mathbb{E}\min(|Z_1 - Z_2|, 1/2) \ge \frac{13}{54} + \frac{1}{8} - \frac{\log 2}{4}.$$

Substitution gives $83/162 + \log 2/3$. Finally, the series

$$\log 2 = 2\sum_{j\ge 0} \frac{3^{-(2j+1)}}{2j+1} < \frac{25}{36}$$

implies the rational bound $241/324 < 3/4$. □

Thus $Q \mid P$ alone is insufficient; full Q^2 alignment is the structural condition behind the simple two-digit law.

11 Security interpretation and limitations

The conclusions of this article are deliberately asymmetric.

(a) The interval inverse relation is efficiently countable and selectable. For a uniform interval secret, the exact posterior can be sampled.

(b) In the two-message deterministic scalar exchange, posterior replacement predicts the peer's key with the honest agreement probability.

(c) The full-range power-of-two experiment has honest agreement at least $c_* > 3/4$.

(d) This agreement bound is distribution-specific and fails for a natural exactly-seven-bit support.

(e) For aligned general moduli $P = JQ^2$, agreement tends to $41/54$; a nonaligned family can instead converge below $3/4$.

None of these statements is an IND-CPA or IND-CCA theorem. No reduction from LWE, LWR, CVP, SVP or another standard hardness assumption is established here. The earlier public CVP attack [2] remains relevant historical cryptanalysis, but the exact counting/posterior route above is a separate arithmetic analysis and does not depend on the correctness of that attack.

The results do not transfer automatically to higher-dimensional or noisy constructions. In particular, additional messages can alter the posterior. Polynomial, Toeplitz/Hankel, and reconciliation-augmented preimage sets are not covered by the one-dimensional floor-sum selection theorem.

12 Reproducibility

The accompanying package contains:

- `A1_SCALAR_MODDIV_v1_1.tex`, the source of this article;
- `verify_a1_v1_1.py`, an arbitrary-precision Python verifier for floor sums, interval counting, small direct protocol checks, upper-half support, and certificate consistency;
- `count_agreement_a1.cpp`, an optional C++17 exact counter for the complete $q = 1, \ldots, 11$ finite closure;
- `check_upper_half_a1.cpp`, an independent direct enumeration of all 134217728 triples in the seven-bit counterexample;
- exact CSV outputs and SHA-256 manifests.

The finite closure is exact integer computation, not Monte Carlo. The analytic range $q \geq 12$ and the asymptotic limits are justified by the proofs above, not by extrapolating finite experiments. The software is evidence for arithmetic implementation and finite cases; it is not a security certification or a proof-assistant certificate.

13 Conclusion

Scalar modulo-division exhibits a useful cautionary combination: exact arithmetic can produce substantial honest agreement while the same deterministic scalar transcript still admits efficient posterior key prediction under a public interval prior. The three-quarters theorem is genuine but distribution-specific; changing the secret support can cross below the threshold. At the same time, the general aligned family has a clean universal asymptotic constant $41/54$, whereas a simple nonaligned family has a limit below $3/4$.

The main methodological lesson is that quotient cryptography must retain the complete joint law of residues, carries, cell positions and secret supports. Agreement constants, inversion properties and security claims are different objects and must be proved separately.

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